

Set 18: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in \$)
------------	--------------	--------------------	---------------

1	Product A	250	20
---	-----------	-----	----

2	Product B	175	15
---	-----------	-----	----

3	Product C	300	18
---	-----------	-----	----

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
library(ggplot2)
```

```
inventory <- data.frame(  
  Product_ID = c(1, 2, 3),  
  Product_Name = c("Product A", "Product B", "Product C"),  
  Category = c("Category 1", "Category 2", "Category 1"),  
  Quantity_Available = c(250, 175, 300),  
  Price = c(20, 15, 18)  
)
```

```
View(inventory)
```

```
# Bar Chart
```

```
ggplot(inventory,  
  aes(x = Product_Name,  
      y = Quantity_Available,  
      fill = Product_Name)) +  
geom_bar(stat = "identity") +  
labs(  
  title = "Product Inventory Quantity",  
  x = "Product Name",  
  y = "Quantity Available"
```

```
) +
```

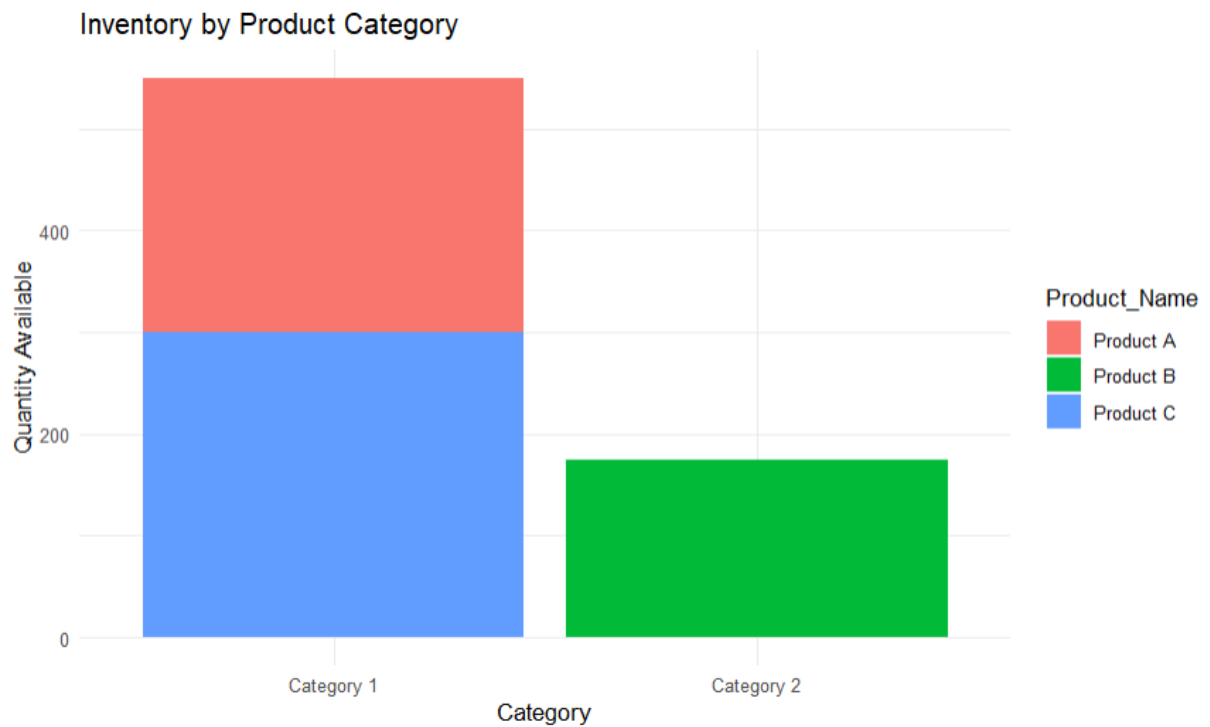
```
theme_minimal()
```



2. Generate a stacked bar chart to show the quantity of each product within different product categories.

```
# Stacked Bar Chart
```

```
ggplot(inventory,  
  aes(x = Category,  
    y = Quantity_Available,  
    fill = Product_Name)) +  
geom_bar(stat = "identity") +  
labs(  
  title = "Inventory by Product Category",  
  x = "Category",  
  y = "Quantity Available"  
) +  
theme_minimal()
```



3. Build a table to show the inventory data for each product. Label the table elements.

View(inventory)

OR

print(inventory)

	Product_ID	Product_Name	Category	Quantity_Available	Price
1	1	Product A	Category 1	250	20
2	2	Product B	Category 2	175	15
3	3	Product C	Category 1	300	18

5 columns

Product_ID <dbl> [1, 3] 0% NA

Product_Name <chr> 3 unique 0% NA

Category <chr> 2 unique 0% NA

Quantity_Available <dbl> [175, 300] 0% NA

Price <dbl> [15, 20] 0% NA

Showing 1 to 3 of 3 entries, 5 total columns

Set 19: Survey Responses Analysis

Dataset: Survey Responses

Survey ID Question 1 Question 2 Question 3

1 A B C

2 B A D

3 C A B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

```
library(ggplot2)
library(reshape2)

survey <- data.frame(
  Survey_ID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

View(survey)

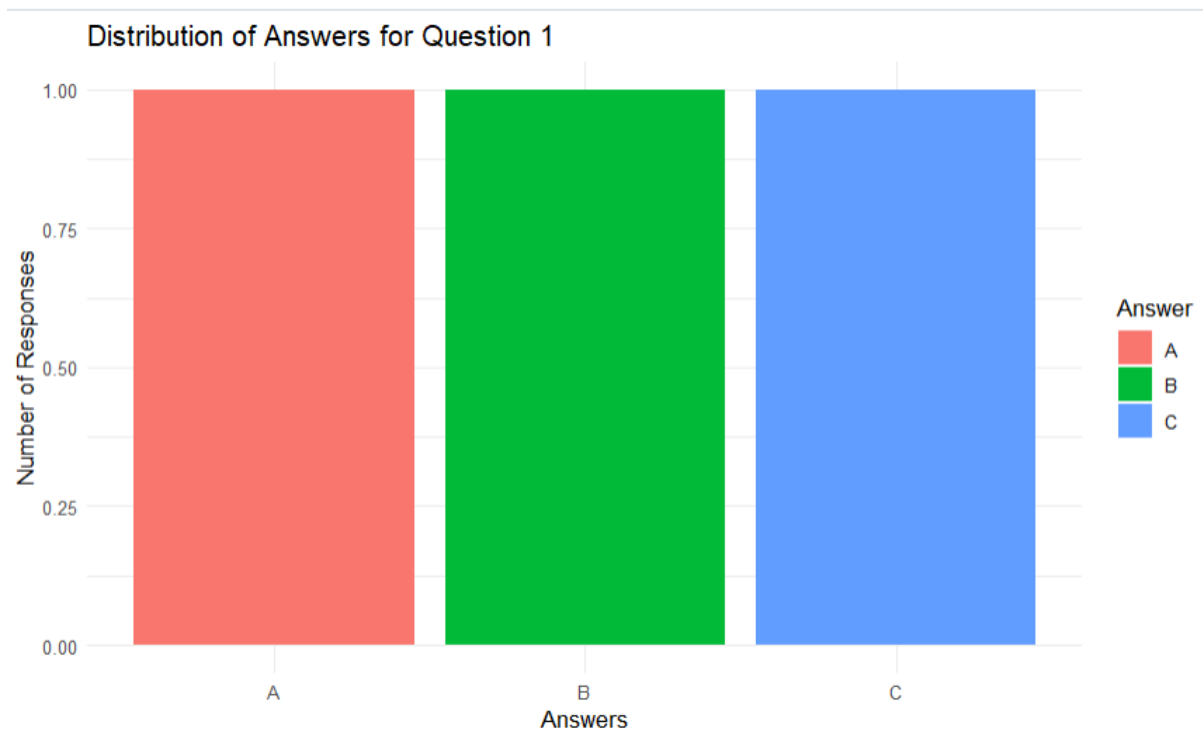
# Grouped Bar Chart

q1 <- as.data.frame(table(survey$Question1))
colnames(q1) <- c("Answer","Count")

ggplot(q1,
  aes(x = Answer,
      y = Count,
      fill = Answer)) +
  geom_bar(stat = "identity",
    position = "dodge") +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Answers",
    y = "Number of Responses"
```

```
) +
```

```
theme_minimal()
```



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

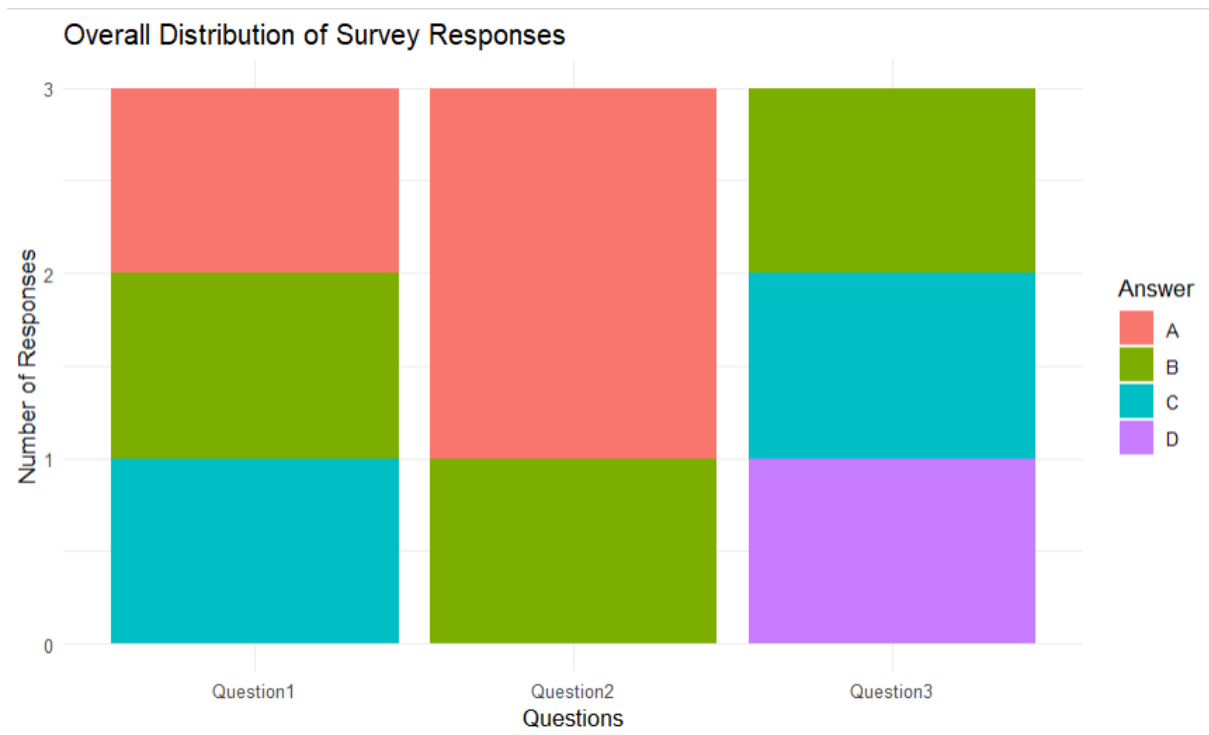
```
# Stacked Bar Chart
```

```
long <- melt(  
  survey,  
  id.vars = "Survey_ID",  
  variable.name = "Question",  
  value.name = "Answer"  
)  
ggplot(long,  
  aes(x = Question,  
      fill = Answer)) +  
  geom_bar() +  
  labs(  
    title = "Overall Distribution of Survey Responses",  
    x = "Questions",
```

```
y = "Number of Responses"
```

```
) +
```

```
theme_minimal()
```



3. Build a table to show the survey response data for each survey. Label the table elements.

```
View(survey)
```

```
# OR
```

```
print(survey)
```

	Survey_ID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

4 columns			
+	Survey_ID	<dbl>	?
[1, 3] 0% NA			
+	Question1	<chr>	?
3 unique 0% NA			
+	Question2	<chr>	?
2 unique 0% NA			
+	Question3	<chr>	?
3 unique 0% NA			

Showing 1 to 3 of 3 entries, 4 total columns

Set 20: Stock Analysis

Dataset: Stock Prices

Date Stock A Stock B Stock C

2023-01-01 100 150 120

2023-01-02 105 152 118

2023-01-03 110 148 122

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock

C) over a specific time period. Label the axes and title the chart.

```
library(ggplot2)
```

```
library(reshape2)
```

```
stock <- data.frame(  
  Date = as.Date(c("2023-01-01",  
                   "2023-01-02",  
                   "2023-01-03")),  
  Stock_A = c(100,105,110),  
  Stock_B = c(150,152,148),  
  Stock_C = c(120,118,122)  
)
```

```
View(stock)
```

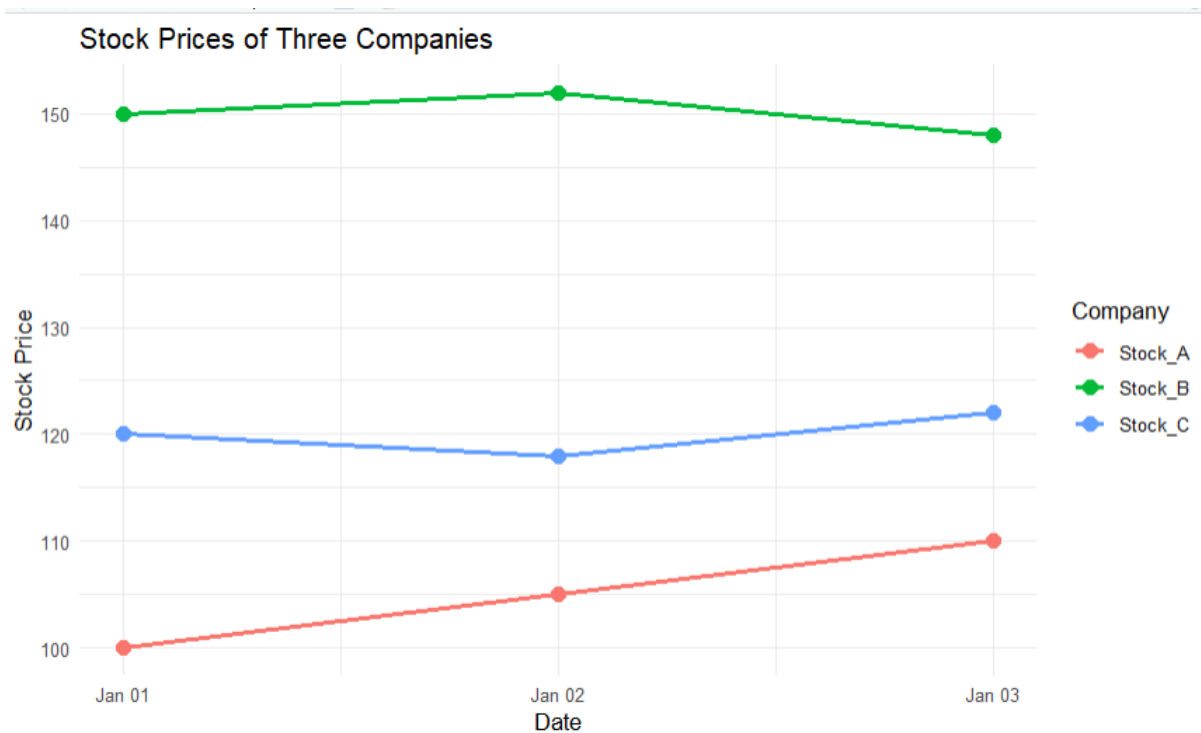
```
# Convert to long format
```

```
stock_long <- melt(  
  stock,  
  id.vars = "Date",  
  variable.name = "Company",  
  value.name = "Price"
```

)

Line Chart

```
ggplot(stock_long,
  aes(x = Date,
      y = Price,
      color = Company,
      group = Company)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(
    title = "Stock Prices of Three Companies",
    x = "Date",
    y = "Stock Price"
  ) +
  theme_minimal()
```



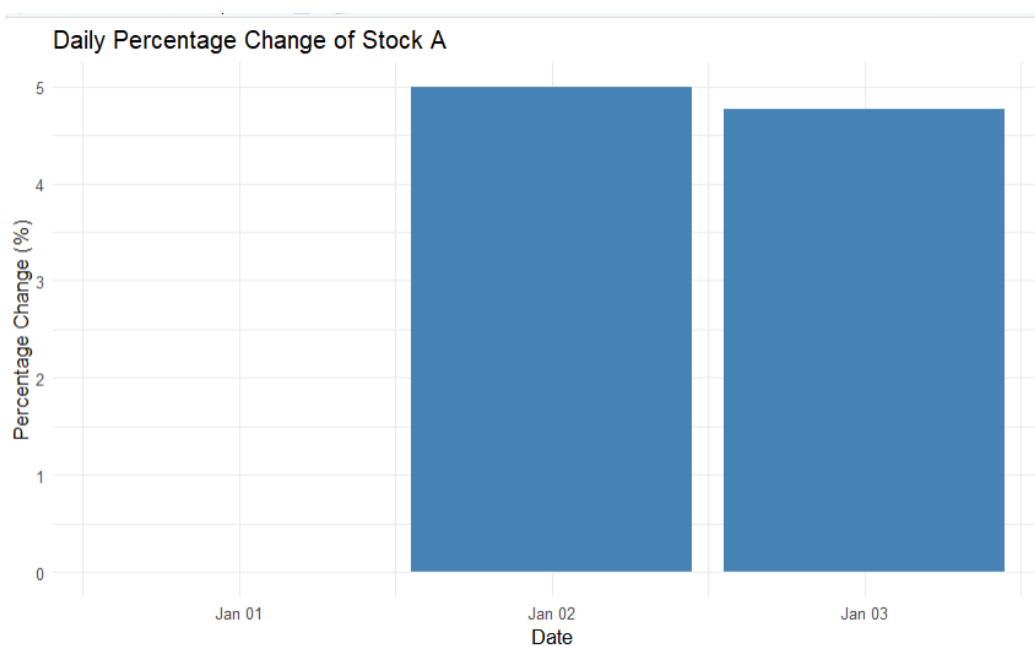
2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.


```

# Percentage Change
stock$Percent_Change <- c(
  0,
  (stock$Stock_A[2]-stock$Stock_A[1])/stock$Stock_A[1]*100,
  (stock$Stock_A[3]-stock$Stock_A[2])/stock$Stock_A[2]*100
)

# Bar Chart
ggplot(stock,
  aes(x = Date,
      y = Percent_Change)) +
  geom_bar(
    stat = "identity",
    fill = "steelblue"
  ) +
  labs(
    title = "Daily Percentage Change of Stock A",
    x = "Date",
    y = "Percentage Change (%)"
  ) +
  theme_minimal()

```



3. Build a table to display the stock price data for each company over the given period. Label the table elements.

View(stock)

OR

print(stock)

	Date	Stock_A	Stock_B	Stock_C	Percent_Change
1	2023-01-01	100	150	120	0.000000
2	2023-01-02	105	152	118	5.000000
3	2023-01-03	110	148	122	4.761905

5 columns				
Date				
[2023-01-01, 2023-01-03]				
0% NA				
Stock_A				
[100, 110]				
0% NA				
Stock_B				
[148, 152]				
0% NA				
Stock_C				
[118, 122]				
0% NA				
Percent_Change				
[0, 5]				
0% NA				

Showing 1 to 3 of 3 entries, 5 total columns

Set 21

Dataset: Energy_Consumption_Data.csv

Sector Region Month Temperature

(°C)

Units_Consumed

(kWh)

Cost

(₹)

Renewable_Usage

(%) Peak_Hours

Residential North Jan 15 320 2100 22 4

Commercial South Jan 24 540 3600 18 6

Industrial West Feb 20 880 5900 12 8

Residential East Feb 18 350 2300 25 5

Commercial North Mar 28 610 4100 20 7

Industrial South Mar 30 920 6200 15 9

1. Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

```
library(ggplot2)
```

```
energy <- data.frame(  
  Sector = c("Residential","Commercial","Industrial",  
            "Residential","Commercial","Industrial"),  
  Region = c("North","South","West",  
            "East","North","South"),  
  Month = c("Jan","Jan","Feb","Feb","Mar","Mar"),  
  Temperature = c(15,24,20,18,28,30),  
  Units_Consumed = c(320,540,880,350,610,920),  
  Cost = c(2100,3600,5900,2300,4100,6200),  
  Renewable_Usage = c(22,18,12,25,20,15),  
  Peak_Hours = c(4,6,8,5,7,9)  
)
```

```
View(energy)
```

```
# Histogram
```

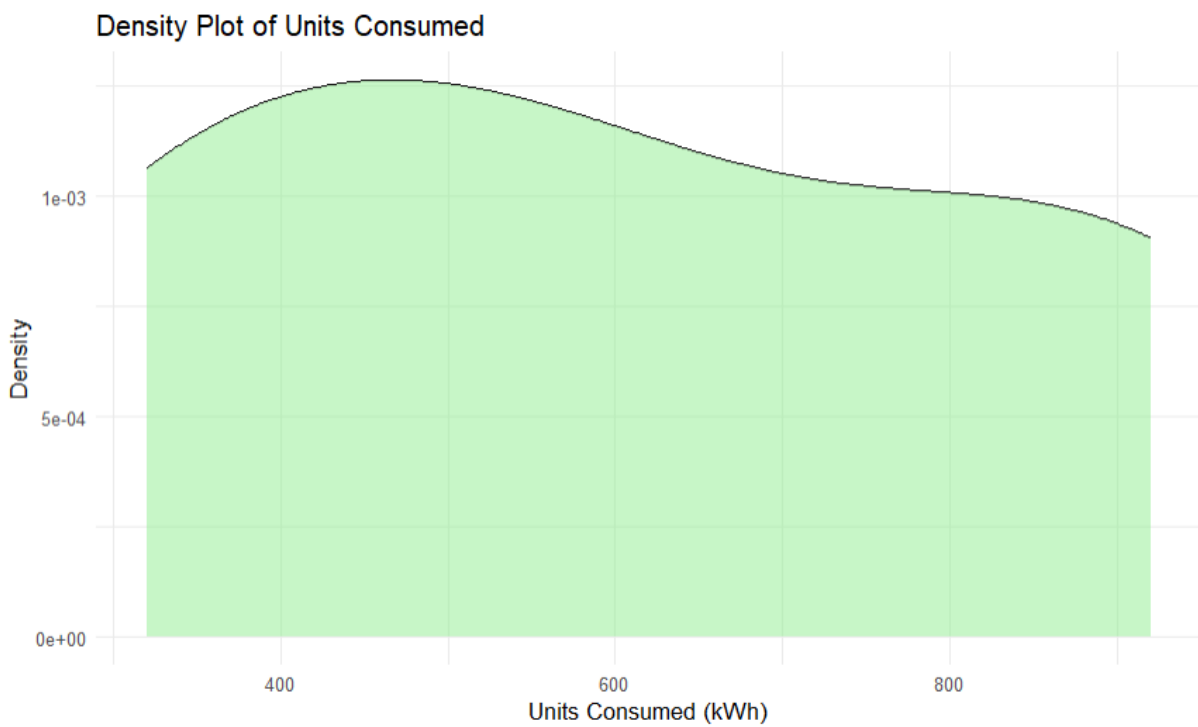
```
ggplot(energy, aes(x = Units_Consumed)) +  
  geom_histogram(binwidth = 100,  
                fill = "skyblue",  
                color = "black") +  
  labs(title = "Histogram of Units Consumed",
```

```

x = "Units Consumed (kWh)",
y = "Frequency") +
theme_minimal()

# Density Plot
ggplot(energy, aes(x = Units_Consumed)) +
  geom_density(fill = "lightgreen",
               alpha = 0.5) +
  labs(title = "Density Plot of Units Consumed",
       x = "Units Consumed (kWh)",
       y = "Density") +
  theme_minimal()

```



2. Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

```

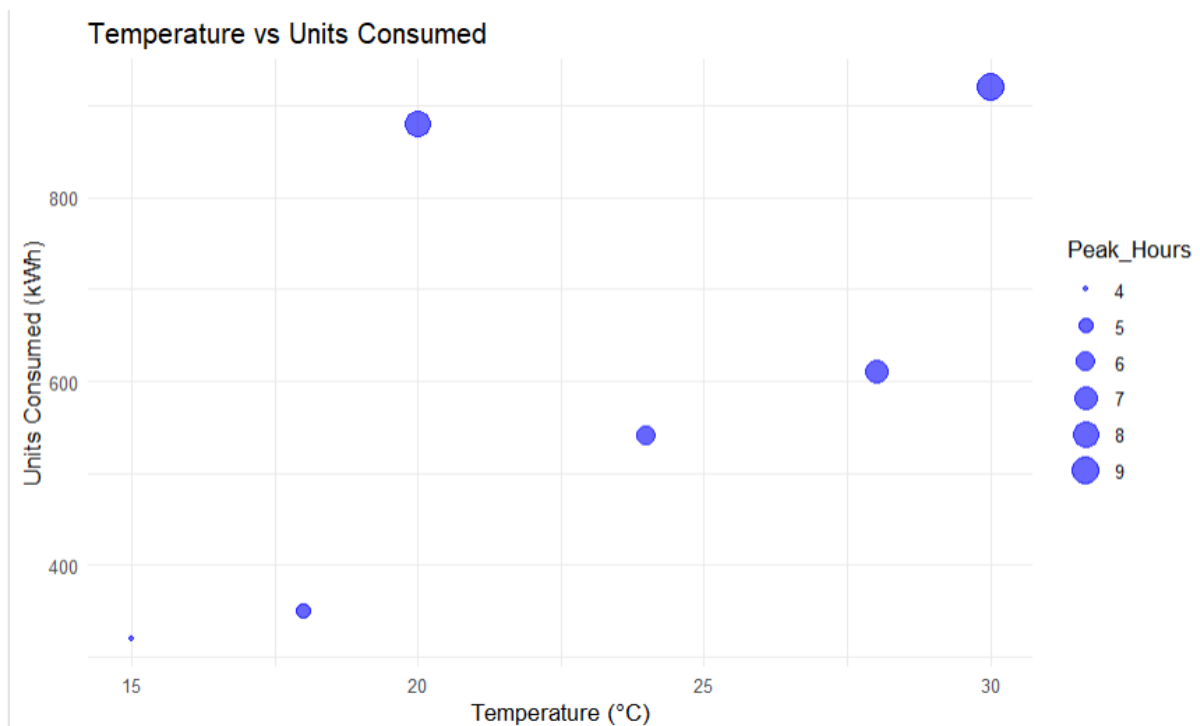
# Bubble Scatter Plot
ggplot(energy,

```

```

aes(x = Temperature,
    y = Units_Consumed,
    size = Peak_Hours)) +
geom_point(color = "blue",
           alpha = 0.6) +
labs(title = "Temperature vs Units Consumed",
     x = "Temperature (°C)",
     y = "Units Consumed (kWh)") +
theme_minimal()

```



3. Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

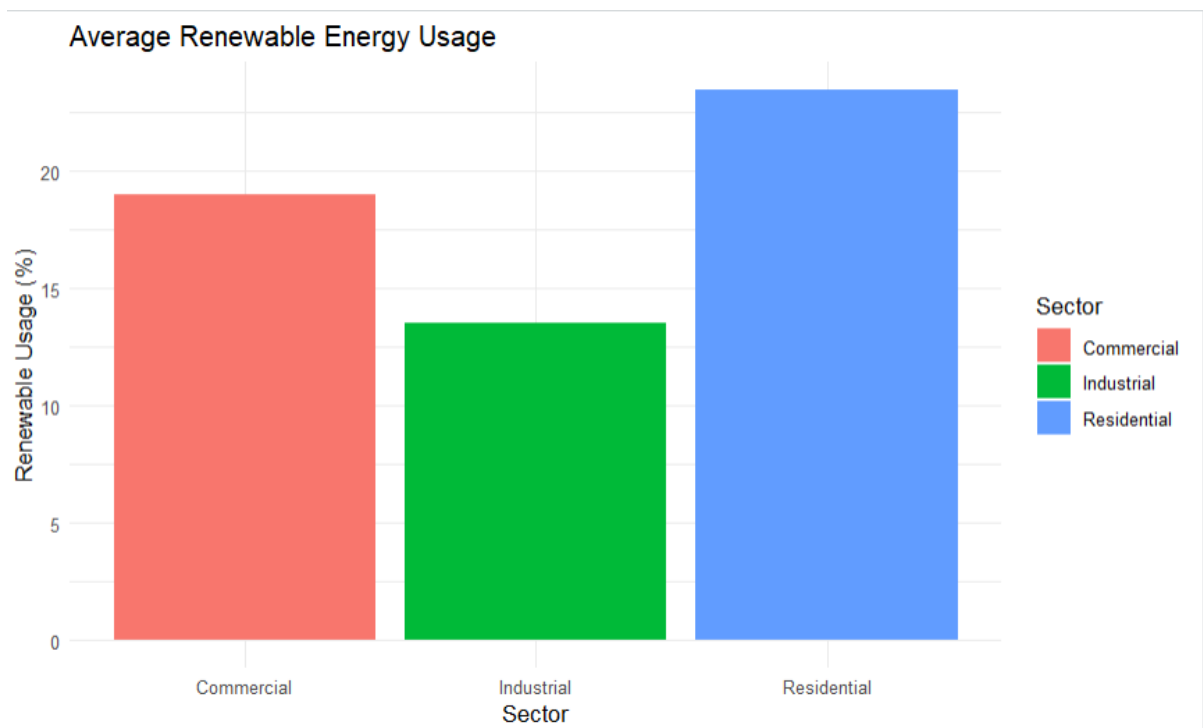
```

# Average Renewable Usage
avg_usage <- aggregate(
  Renewable_Usage ~ Sector,
  data = energy,
  mean
)

```

```
# Bar Chart
```

```
ggplot(avg_usage,  
  aes(x = Sector,  
      y = Renewable_Usage,  
      fill = Sector)) +  
geom_bar(stat = "identity") +  
labs(title = "Average Renewable Energy Usage",  
  x = "Sector",  
  y = "Renewable Usage (%)") +  
theme_minimal()
```



Set 22: Time Series Analysis

Dataset: Monthly Sales Data

Month Sales (in \$)

January 15000

February 18000

March 22000

April 20000

May 23000

1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

```
library(ggplot2)
```

```
sales <- data.frame(
```

```
  Month = factor(c("January", "February", "March", "April", "May"),
```

```
    levels = c("January", "February", "March", "April", "May")),
```

```
  Sales = c(15000, 18000, 22000, 20000, 23000),
```

```
  Advertising_Budget = c(2000, 2500, 3000, 2800, 3200),
```

```
  Product = c("Product A", "Product B", "Product A", "Product C", "Product B")
```

```
)
```

```
View(sales)
```

```
# Line Chart
```

```
ggplot(sales,
```

```
  aes(x = Month,
```

```
    y = Sales,
```

```
    group = 1)) +
```

```
geom_line(color = "blue", linewidth = 1) +
```

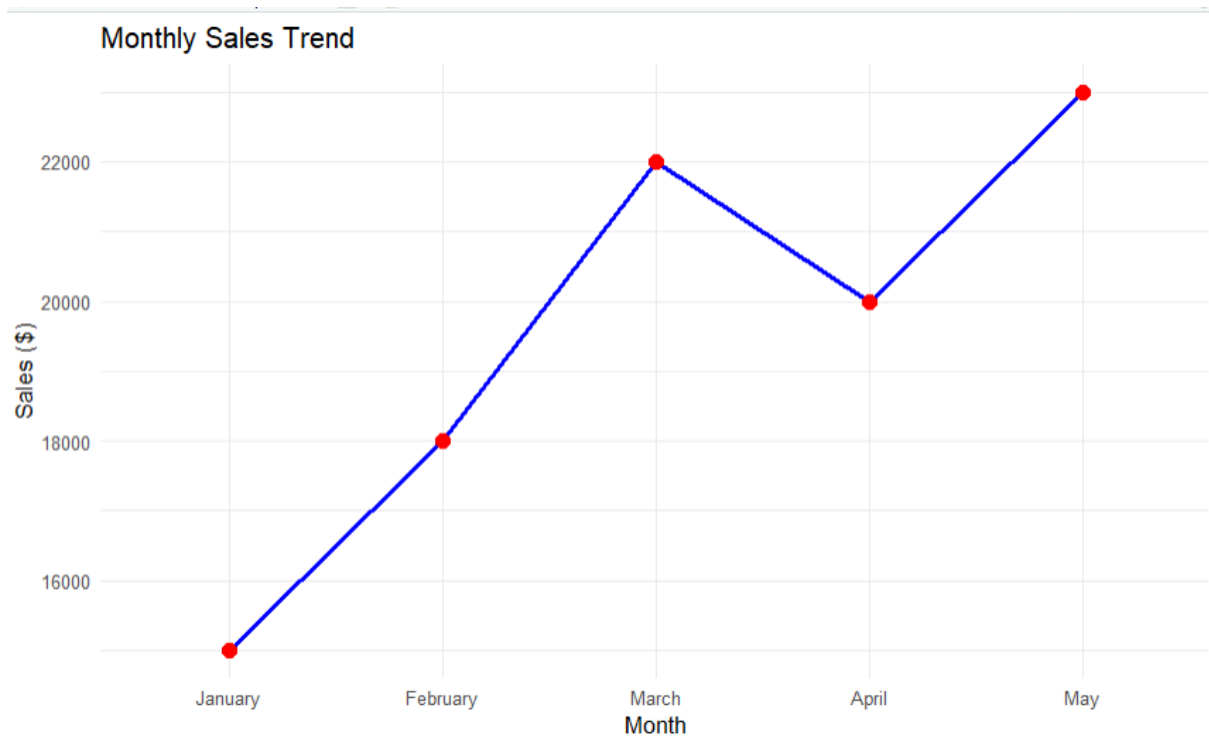
```
geom_point(color = "red", size = 3) +
```

```
labs(title = "Monthly Sales Trend",
```

```
  x = "Month",
```

```
  y = "Sales ($)") +
```

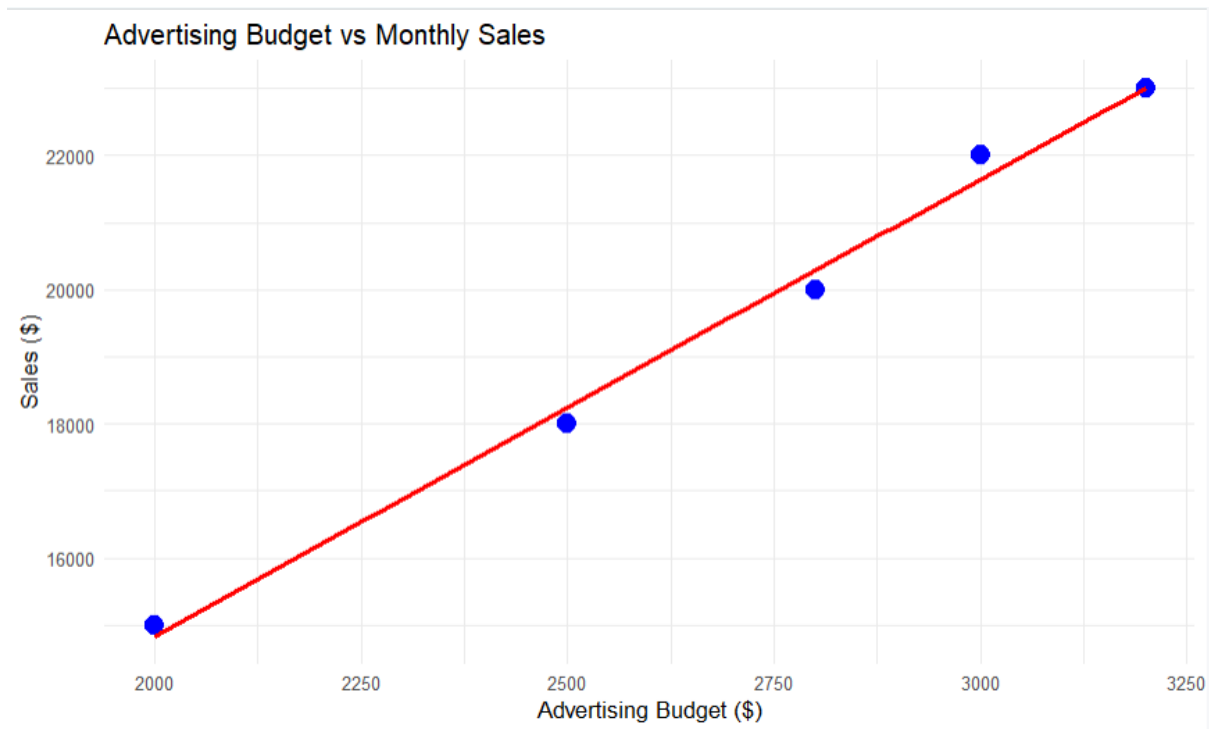
```
theme_minimal()
```



2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

Scatter Plot

```
ggplot(sales,  
  aes(x = Advertising_Budget,  
    y = Sales)) +  
  geom_point(size = 4,  
    color = "blue") +  
  geom_smooth(method = "lm",  
    se = FALSE,  
    color = "red") +  
  labs(title = "Advertising Budget vs Monthly Sales",  
    x = "Advertising Budget ($)",  
    y = "Sales ($)") +  
  theme_minimal()
```

4. In R, create an autocorrelation plot to identify seasonality in the time series data.

```
# Autocorrelation Plot
```

```
sales_ts <- ts(sales$Sales, frequency = 12)
```

```
acf(sales_ts,
```

```
  main = "Autocorrelation Plot of Monthly Sales")
```

